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PEE 524 — Past Questions & Materials

by Renance

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Paper 1 of 1 — PEE 524 INTEGRATED FIELD DEVELOPMENT AND MANAGEMENT Lecture Notes

Level 500 • Second Semester • Handout

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INTEGRATED FIELD DEVELOPMENT & PLANNING (PNG 517)

COURSE CONTENT:

Introduction: [stage-gate process of oil field development, definitions of field development planning (FDP) and the keywords]. Stages and sections of FDP. Criteria considered in field development. SWOT analysis during field development. Conditions for and barriers to field development. Methods of improving field development performance. Field development cycle. Factors affecting options employed in field development. Subsurface and surface aspects of FDP. Production system options for offshore field development. Gas field development. Marginal field development. Approaches to marginal field development (farm out/farm in arrangement). Production system options for marginal field development. Management of marginal field. New technologies employed in field development.

Definitions of types of Reserves. Method of reserves estimation (deterministic and stochastic modeling's); volumetric, performance history evaluation (decline curve analysis and material balance), reservoir simulation, etc.

Definitions of project planning and the keywords. Reasons for planning. Planning horizons. Planning approaches. Criteria for effective planning. Steps in planning. External factors affecting project planning.

Definitions of project management and the keywords. Basic principles/conditions for project management. Process flow in project management. Reasons for project management.

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INTRODUCTION

Throughout its life cycle for exploration to abandonment, an oil field requires the participation of many different specialists, working in teams, to make major investments and operation decisions. These decisions must be made under conditions where reservoir data is obtained from limited sources and uncertainties exist not only about the costs and future performances of the assets, but also with respect to the future price, market demand for the produced oil and gas, the advances in oil field technology and computing, and the smaller and poor-quality reservoirs and are often discovered.

Because of the risks, the complexities and the large investments needed to develop an oil field, most companies divide the analysis and decisions into discrete, sequential stages using a disciplined, project management process often referred to as a 'stage-gate' process, shown in figure 1 below.

STAGE 1 STAGE 2 STAGE 3 STAGE 4 STAGE 5

Optimize Execute field Manage
 Identify and Analyze production life
 preferred development
 assess alternative cycle
 development FID plan
 exploration Proceed development Proceed Proceed
 plan
 opportunities options
 IAEO AADO OPDP EFDP MPLC
 FDP

Figure 1: the stage-gate process for field development

The stage-gate process for field development shown above is typical of those used by the petroleum industry as a project progresses from concept to production

There are five stages in oil field development:

Stage 1 is devoted to identification and assessment of opportunities. This is essentially the exploration process.

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Stage 2 and 3 are the development planning stages where many teams of specialists perform feasibility studies in order to define the optimal development plan for the field including the schedule and project economics with the aim of obtaining a final eminent decision (FID).

Preparation of an optimal field development plan (FDP) is an interactive process, in which concepts from disciplines are combined into a general plan for evaluation. For example, when a concept considered to be optimal from a reservoir management perspective is placed in the total scheme, it could make project economics less favorable. Individually optimized FDP components from various disciplines would not likely combine into an optimized overall FDP.

After the FID is made, stage 4 involves the execution of the FDP. This is essentially the field construction period.

Stage 5 is dedicated to the long-term management of production operations. At this state, the objectives are to optimize production, improve efficiency and maximize the net present worth of the asset by often making additional investments to recover incremental reserves.

Within each stage, teams of specialists shown in the lower part of figure 2, perform (activities) studies, prepare project documents and then present their findings with recommendations as appropriate to an executive decision board to move the project to the next stage. The board may agree to move the opportunity forward, ask the team to do more studies, put the opportunity on hold or terminate its consideration altogether. The review sequel to FDP execution is called After-Action Review (AAR).

Executive Review Board

Presentation

PAR Proceed

to next

Activities

stage

Documents

PAR= Post Activity Review

Project manager/team leaders work teams

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Gate-stage process for field development (in figure 1) could also be presented as the field development procedures shown below in figure 3.

FDP

IPES CRS FDP FID

Execution

PAR I PAR II PAR III AAR

Figure 3: felid development procedures

IPES = Integrated Petroleum Engineering Studies

PAR I = Post Activity Review I (For Possible Amendments)

CSR = Concept Selection Review

PAR II = Post Activity Review II (for possible amendments)

FDP = Field Development Planning

PAR III = Post Activity Review III (for possible amendments)

FID = Final Investment Decision

FDP Execution = Field Development Planning Execution

AAR = After-Action Review (for possible amendments)

IPES: Integrated Petroleum Engineering Studies

The studies are integrated because it involves geological studies, petrophysical studies, reservoir engineering studies and production engineering studies.

CSR: Concept Selection Review

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Here, the development concepts are selected and reviewed. For example, surface development concepts reviewed entails whether the field production should stand above or be combined with that of another field. Sub surface development concepts review entails whether the wells should be vertical, deviated/inclined, or horizontal. For offshore development, the choice of subsea completing, floating platform and storage offloading (FPSO), and number of separators should be reviewed.

FDP: Field Development Planning

This explains WHAT will be done in terms of exploitation, design bases, engineering, operations and maintenance, and economics (EDEOE).

ID: Final Investment Decision

These decisions are taken on the primary goal of field development i.e. minimizing profit at a minimized investment cost.

FDP Execution: Field Development Planning Execution

This is the actual implementation of the stages involved in field development.

The project leader is appointed to coordinate, schedule and steward plan preparation. A clear definition and allocation of responsibilities among team members helps to ensure that the plan is prepared in an efficient and timely manner. Also needed is definition of approval and review lining and authority by management. Periodic management review and approval of the interim status and results of the plan is important.

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DEFINITION OF FDP AND THE KEYWORDS

Field Development Planning

Field development planning is a multi-disciplinary and integrated approach of performing and analyzing feasibility studies on the exploitation, design basis, engineering, operations and maintenance, and economics aspects of developing an oil field in order to define the optimal development plan and obtain a final investment decision.

The multi-disciplinary project team usually needed in field development planning includes geologic and geophysical, engineering, operations and management personnel. Field development planning explains WHAT will be done whereas the project execution plan explains HOW it will be done.

Marginal Field Development Planning

marginal field can be defined as discovery (appraisal or not) in the range of 5 to 20 million barrels reserve, that has remained undeveloped for a fairly long period of time (usually 5 to 10 years) and fall on the border line between economic and uneconomic reserves hence they are economically sensitive.

Reasons for economic sensitivity of marginal field development

- i. Field size to field ranking
- ii. low reserves either as a result of low STOIP or low recovery factor
- iii. unfavourable fiscal regime
- iv. marginal economics which could either be as a result of low crude oil price or low gas price or both.
- v. low production rate especially in thin reserves
- vi. relatively long distance from existing infrastructure

vii. water depth

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viii. reservoir rock and fluid characteristics such as low API gravity, high viscosity, low porosity, low permeability k.

ix. high and unfavourable development cost. (it must be noted that the development of some field that are uneconomical in some parts of the world (such as West Africa) due to high development cost may be profitable in some parts of the world (such as Gulf of Mexico) where they have low development cost.

Approaches to Marginal Field Development

Several approaches are open to the development of marginal field. the most appropriate development scheme/approach will, however, depend on how innovative the arrangement is and how favourable the prevailing fiscal regime is.

The base approach to the operations of marginal field is the farm out/farm in arrangement

Farm out/farm in arrangement

In this type of arrangement, the concessionaire is not interested in drilling at a time but agrees to assign the lease of the field or a portion of it to another operator or farminee who is ready to develop the acreage. He may not retain an interest (royalty) in the production. This type of arrangement is characterized by:

- i. Interest acquisition: in this case, the farminee carries out an agreed work programme in return for a percentage interest from the field. He will, however, take responsibility for all aspects of development including drilling, operation and finance.
- ii. Overriding Royalty: in this case, the farm out party assigns his interest to the farminee who will earn an accruing interest. However, after field development when the farminee must have earned an agreed payment on the investment, the farminee will back in on agreed rate of interest.
- iii. Tariff Payment: in this case, the farminee receives a fixed or adjustable tariff in dollar per barrel. The fee could also come as a percentage of net production.
- iv. Others in such as acreage swap, partial relinquishment, transfer of as taken interest in the asset, sublease or subcontract option, etc.

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Production System Options For Marginal Field Development

- i. existing production platform
- ii. floating production system
- iii. mobile offshore production unit (MPU)
- iv. subsea completion system
- v. small facilities, unmarinea structures or light-weight platform

Management of marginal field

The need for higher return on marginal investment forces marginal field operators to look into ways of carrying out petroleum business different from other methods. Some of the new options/alternatives being adopted include:

- i. Partnership and allowance concept, ii. Common sharing of production facilities and iii. use of total quality management (TQM)

Risk and uncertainties are inherent in petroleum investment. They must be identified and the risk must be quantified. Dealing with risk and uncertainties way affects investment strategy. Some of the current fiscal regime that will impart marginal field economy include royalty, signature bonus, production bonus. the property varies from 85% to 75%, investment task credit 10%

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OIL FIELD

An oil and gas field is an area consisting of a single or multiple hydrocarbon reservoirs grouped or related to the same structural features or stratigraphic condition.

Or an area defined to be a field by the government or regulating agency.

Reservoir:

A petroleum reservoir is a porous and permeable formation containing an individual or separate natural accumulation of producible hydrocarbons and other substances which is confined by an impermeable rock or water barrier and is characterized by a single natural pressure system.

Development:

Oil fields are developed because of economic incentives, domestic needs, national concerns and global incentives.

Development of a field that has no history of initial production is called green field development. However, brown field development indicates that the field has history of production prior to the new development.

Planning is a rational approach to reach predetermined objectives. The purpose of every plan and supporting plans is to facilitate the accomplishment of organization objectives. It involves selecting missions and goal and determining the set of action needed to achieve them. It involves deciding in advance what to do, how to do it, when to do it, where to do it and what are needed to do it.

STAGES AND SECTIONS OF FDP

Stages in field development planning

- a. Analysis of Alternative Development Options
- b. Optimization of Preferred Development Plans

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Sections of field development planning

- a. Exploitation
- b. Design basis
- c. Engineering
- d. Operations and maintenance
- e. Economic

Exploitation

The exploitation is the most important part of a field development planning because it contains the subsurface plan for the development; the in-place volumes, reserves, field deliverability, and reservoir management strategy determines the type of completion, facilities and transportation schemes.

The uncertainties in determining reserves and deliverability from a limited number of exploration/deliberation wells make the conclusions presented in the exploitation section to be the least reliable composed of field development planning. Hence, considerable amount of time and energy are spent in preparing the exploitation section and the risks in the reserves and rate predications obtained as well assessed and weighed.

There is the need for the development planning process to start during the prospect evaluation phase of an exploration programme because of the significant of the exploitation section. This is with a view to maximizing the value of information obtained from each well and avoiding post discovery data requirement planning.

Some of the key elements of the exploitation section are:

- i. Geology and geophysics
- ii. Estimation of hydrocarbons in place
- iii. Reserves estimation
- iv. Reservoir management
- v. Completion and work over

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- vi. Reservoir and production engineering
- vii. Development drilling
- viii. Well performance prediction
- ix. Production and injection profile
- x. Gas production profiles

Design Basis

The design basis section summarizes the major parameters of development planning and includes a vivid description of the aims and objectives of the field development planning. The major parameters are employed in the engineering section to arrive at detailed facilities and equipment requirements and specifications.

Some important components of the design basis are:

- i. Reservoir and fluid characteristics
- ii. Production and fluid characteristics
- iii. Wells
- iv. Engineering, operating and maintenance philosophy
- v. Installations
- vi. Transportations
- vii. Infrastructure
- viii. Government and third party consent
- ix. Safety and environment

Engineering

The detailed engineering used to design the production facility and transportation systems are contained in the engineering section. This section gives the details of the methods used, major assumptions and options (or alternatives) investigated.

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Project schedules in the engineering section should be presented in the early stage of project particularly prior to management or government approval because detailed design engineering would not have completed.

Some of the key elements of the engineering section are:

- i. Field installation and drilling facilities
- ii. Production facilities
- iii. Hydrocarbon transport system
- iv. Project organization

Operations and maintenance

The major parts of the operations and maintenance section are organizational construction, operating and maintenance needs, safety and emergency planning and provision. As data is collected during the operational phase of field development, the exploitation plan should be re-assessed to implement necessary changes; hence, feedback loop is another element of the operations and maintenance plans.

Some of the key elements of the operation and maintenance section are:

- i. Production operations
- ii. Maintenance operations
- iii. Drilling, completion and surveillance

Economics

Cost and price forecasts are combined with output from the previous sections to permit management evaluation of project viability; this is done in the economics section of the development. Risk analysis and associated sensitivity studies are important parts of this section.

Critical factors that could enhance or jeopardize the project must be identified, quantified, and their probabilities discussed.

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Project Planning

Definition of Project

A project is a related set of activities having a discrete beginning and an end, and resulting into delivery of specific product or service.

But an ongoing business or maintenance operation is not a project. A project exists only after a decision had been made to address a business need. Funding is added to support the execution and measurable goals and objectives are well defined.

Definition of Planning

Planning is a rational approach to reach predetermined objectives. The purpose of every plan and supporting plans is to facilitate the accomplishment of organization objectives. It involves selecting mission and goals, and determining the set of actions needed to achieve them. It involves deciding in advance what to do, how to do it, when to do it where to do it, who to do it and what are needed to do it.

Reasons for planning

- i. to focus on objectives
- ii. to prepare for unforeseen circumstances
- iii. to offset uncertainties unexpected changes
- iv. to make proactive rather than reactive measures on decisions
- v. to minimize cost of operations and to facilitate control

Planning Horizons

Major planning horizons are:

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- i. Long-Range Planning (strategic planning): the long-range planning system of an organization elaborates an effort to improve overall effectiveness. if focuses on a long-term horizon of more than a year, roughly about 5 years and over.
- ii. Medium-Range Planning (Tactical Planning): Usually it is a plan for 1 year, though some companies may consider 1 to 3 years.
- iii. Short-Range Planning: it is a plan for just 3 to 6 months. (operational planning)

About

Or over

Strategic objectives 5 years Strategic plannings

6-12

months or

Tactical objectives more Tactical planning

0-6

months

Operational objectives Operational planning

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PROJECT MANAGEMENT

Definition of Project

A project is a related set of activities having a discrete beginning and an end, and resulting into delivery of specific product or service.

But an ongoing business or maintenance operation is not a project. A project exists only after a decision and been made to address a specific business need. Finding is added to import and execution and measurable goals and objectives are well defined.

Definition of Management

Management is usually defined as a process of designing and maintaining an environment in which individual working together in groups accomplish efficiently with selected aims and objectives.

Project management definition

In recent times, project management has been defined as getting THINGS done PROPERLY through PEOPLE on TIME and a BUDGET. This has helped to effectively define and establish the basic principles or certain basic conditions for project management.

Basic principles/conditions for project management

Things: identification of needs or work objective and this is defined in terms of result required

Properly: following laid down standard and establishing control system for quality assurance

People: team is required and the people in project team must be specified and well defined.

Time: a realistic schedule must be established

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Budget: a financial plan on the project must be drawn up and ensure that work objective is carried out.

Process flow in project management

1. Identification of needs
2. Project initiation
3. Project planning
4. Project execution
5. Project control
6. Closing out the project

Reasons for project management

1. To achieve overall mission and objectives in an organization
2. It helps management to understand the status and scope of business activities to implement
3. To enable an organization to carry out a project in discipline, well managed and competent manner. The project is successful when. It is of good quality, completed in time, within budget and is safe i.e. accident free.

Integrated skills needed for project management

1. Scope management
2. Quality management
3. Time management

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4. Cost management
5. Risk management
6. Procurement management
7. Human resource management
8. Communication management
9. Integration management

Ten Commandments of Project Management

1. Thou shall narrow project scope
2. Thou shall not allow a fat project team
3. Thou shall empower project team
4. Thou shall request for full time commitment
5. Thou shall not provoke work fatigue/burn out
6. Thou shall not tolerate quick and dirty works
7. Thou shall reward hard work with success
8. Thou shall have project review panel
9. Thou shall seek outside assistance as needed
10. Thou shall use project management tools

We considered economical

Some of the key elements in the economic section are:

- i. Capital cost estimation
- ii. Operating cost estimation
- iii. Revenues
- iv. Overall economics

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CRITERIA CONSIDERED IN FIELD DEVELOPMENT

- a. Technical Consideration: these include field location (land or sea) field size, multiple reservoirs, ultimate recovery, reservoir attributes, crude quality STOIP, and separator capacity.
- b. Economic Considerations: these include crude oil price, economic policies, field expenses, capital cost, market control and constraints, taxation, project viability, demand for oil, royalties.
- c. Commercial Considerations: these include capital availability, inflation, interest rate, lease life, contract term, financing option, (i.e. lease loan, equity), field running policy.
- d. Operational Consideration: these include field location (land or sea), infrastructure, power source, equipment maintenance, operational cost, (e.g. staff remuneration) field accessibility, staff accommodation and feeding, telephone.
- e. Political Consideration: These include security, international boundaries, and boundaries with fields belonging to other companies, government policies, community issues, technology, and manpower availability. OHSE issues.

SWOT ANALYSIS OF FIELD DEVELOPMENT

SWOT analysis is a strategic planning method used to evaluate the strength, weakness, opportunities, and threats involved in a project or business venture. It involves specifying the objectives of the business venture or project and identifying the internal and external factors that are favorable and unfavourable to achieve that objective.

- Strength: characteristics of the business or team that give it an advantage over others.
- Weakness (or limitations): are characteristics that place the team at a disadvantage relative to others.

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- Opportunities: external chances to improve performance (e.g. make greater profits) in the environment.
- Threats: external elements in the environment that would cause trouble for the business or project.

Identification of SWOTs is essential because subsequent steps in the process of planning for achievement of the selected objective may be derived from the SWOTs.

SWOTs analysis is sometimes called SWOTs matrix because it consists of four elements in a 2x2 matrix, as shown below.

Advantages of SWOT Analysis

Helpful to achieving the Harmful to achieving the objectives objectives 1. it is a source of information for strategic planning

Internal origin

2. builds organizations strength

Strength Weakness 3. reverse its weakness

4. maximize its response to opportunities

5. Overcome organization external origin

Opportunities threats, etc.

Threats

SWOTs analysis is instrumental in strategy formulation and selection. It is best used

Advantages as a guide,

of SWOTs Analysis

and not as a prescription. 1. it is a source of information for strategic planning

2. builds organisations strength

3. reverse its weakness

4. maximize its response to opportunities

5. overcome organization threats

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etc.

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Example of SWOT Analysis of Field Development

Strengths Weakness

□ sweet oil □ sour gas

□ high API (i.e. low gravity) □ low API (viscous oil)

□ high STOIP □ low STOIP

□ high FGIP □ low UR

□ Ageing facilities

Opportunities Threat

□ high oil price □ vandalism

□ high gas price □ community problems

□ advanced technology □ theft of equipments

□ gas reserves

Conditions for successful field development

i. Field development completion within budget

ii. Field development completion at the most competitive cost

iii. Field development completion on/ahead of schedule

iv. Field development completion within the shortest cycle time

v. Use of optimum technology (i.e. cost effective, easy start-up operation and maintenance)

- vi. Facilities and equipment satisfying the intended objectives
- vii. Sage and operable facilities and equipment
- viii. Good records of quality, health, safety and environmental (QHSE) issues

Barriers to Successful Field Development

- i. Change in scope when objective changes
- ii. Insufficient planning time
- iii. unrealistic objectives
- iv. Contractor's poor performance
- v. Forces outside one's control (e.g. militant actions or explosion)

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Methods of improving field development performance

- i. Bench marking with best practices (compare your performance with those of outsiders)
- ii. Identify performance gaps (i.e. gaps between expectations and actual)
- iii. Develop improvement plans
- iv. Establish control and monitoring systems

Field Development Cycle

Exploration

Development Option

Appraisal

Safety

Feasibility Cost Environment

Reservoir Modeling

Operations philosophy

Finance

Economic Robust

Monitoring Politics

Development plan (Optimizing preferred

Development Activities option)

Programme Discussions

Budget proposal

Factors Affecting Options Employed in Field Development

A good development option will generally aim at reducing cost of development while conserving capital and minimizing financial risk. Such an option must optimize production and cash flow. The best option for a particular field will depend on several factors such as:

- i. Field location (onshore or off-shore)

ii. Estimated reserves and ultimate recovery

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iii. Production rate

iv. distance to existing infrastructure

v. fieldlife

vi. availability of existing equipment

vii. finance

viii. environmental conditions

Surface Aspects of Field Development Plan (or subsurface realization)

A good subsurface development plan will contain at least the following:

1. History and Status: * discovery and appraisal

* Licenses and operators involved

* Offset operators and local infrastructure

* Offset and look-like field comparison

2. Geology and Geophysics: * Seismic data

* Structural interpretation

* Stratigraphical and offset well data

* Reservoir development of the field

* Maps and diagrams

3. Petrophysical and Reservoir (Fluid properties): * petrophysical interpretation

* routine core analysis at atmospheric

and overburden pressure and

temperature

* special core analysis- cap-pressure,

relative permeability, rock

compaction

* PVT analysis

* reservoir fluid properties analysis

4. Reserves: * Hydrocarbons in place

* Hydrocarbons recovery plan

5. Reservoir Management: * geologic and engineering models of reservoir

* drive mechanisms and reservoir continuity

* well and field deliverability

* well planning, well spacing and completion strategy

* reservoir development plan

* development drilling plan-timing and location

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- * pressure maintenance scheme
- * artificial lift
- * improved recovery
- * performance measurement
- * oil, gas and water production forecast
- * reservoir management policy

Surface Aspects of Field Development Plan (i.e. surface facilities options)

The scope of surface development covers all the activities required to bring the hydrocarbons from the wellhead to the point at which custody is transferred to the customers. There are several technical tasks that a surface facility engineer must perform in order to bring hydrocarbon from. The wellhead to the customers, satisfy his own operating needs and dispose off the waste product; the activities involved are:

i. Technical Scope of Surface Facilities

Transfer

wellhead Gathering Treat/Condition Pumping Store (Through
i.e. Separation, (and pipeline and
dehydration, compression measure)
sweetening)

Customers

ii. Functional analysis

Design Construct Operate Maintain Modify

iii. Managerial Activities

Production System Options for Offshore Field Development

- i. production platform (PP): The most commonly used production type include fixed platform, semi-submersibles, monohaul production vessels, tension-leg platform (TLP)
 - ii. Floating Production Systems (FPS): these include tension-leg platforms, tankers, barges, semi-submersible. floating production systems are proven technology in water
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depth between 800 to 900m. The main components of a floating production system include top-side facilities, hull, station keeping system, risers, sub-sea wells, flow lines and offloading systems.

- iii. Mobile Offshore Production Unit (MOPU): this is a mobile drilling rig that has been converted to a production facility by replacing the drilling equipment with customized production system. It can be leased and therefore suitable for a discovery that is too small to justify the cost of a fixed platform.

A MOPU can even be used to produce and evaluate a reservoir or field before committing to a platform design.

- iv. Floating Production Storage and Offloading (FPSO): although it appears as if

floating, it is somehow fixed.

v. Subsea Completion System (SCS): Subsea production systems have gained wide acceptance in areas where the water depth is diver reachable (i.e. can be reached by divers). It is recognized as a cost effective way of bringing on marginal fields.

A subsea system consists of subsea template, free-standing production riser, subsea trees, subsea well control systems.

Different arrangements possible are (i) a subsea well producing to existing facilities, (ii) a subsea well producing to semi-submersible, (iii) a subsea well producing to tankers and bargers.

Gas Field Development

Gas field development is very similar to oil field development in terms of technical requirement details.

there are three major differences between oil and gas FD:

- i. Gas field development requires a lot of investment on infrastructure; the biggest facility and equipment being pipelines and compressors,
- ii. Gas Field Development involves liquefied natural gas (LNG) or gas to liquid (GTL) plants

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- iii. Economic expenses on investment on gas field development can only be recovered in the long run compared to the short-run recovery of expenses on investment on oil field displacement.

Planning approach

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Top-down-approach:

ADVANTAGES:

1. Quick decision making
2. It saves time
3. Proper management of project
4. Management of staff are in the best position to decide what in a project
5. It is good for emergency planning

DISADVANTAGES

1. Lower staffs will not be productive and creative
2. Lower staff are not consulted, so execution of plan will be difficult
3. Rebellion and industrial strike.

BOTTOM-UP APPROACH

ADVANTAGES

1. Plan is more realistic
2. There is room for creativity
3. It encourages wider participation in planning

DISADVANTAGES

1. Decision making will take time
2. Conflict of interest – management might not adhere or reckon with goals of the (technicians) lower workers.
3. It is not cost effective and no proper management of resources
4. Each section wants to exaggerate their need

GOALS DOWN-PLAN UP APPROACH. (SYNERGY OF BOTH APPROACHES)

ADVANTAGES

1. Effective planning
2. Encourages management – lower ranks workers interaction

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3. Prevent conflict
4. Sense of belonging
5. Vision

DISADVANTAGES

1. Time wasting
2. It does not give room for emergency planning
3. It could be a manipulating tool to “cheat” workers.

Planning Approaches

Three basic approaches are

1. Top-down approach: this planning approach is based on the principle that it is only the top management that has the means and knowledge for effective planning for an organization. Therefore, all the planning activities are carried out at the top. This model has been borrowed from military organizations where the generals prepare the plans and i.e. lower officers and ranks carry them out. In business organizations, this approach is based on a theory of management which states that "employees dislike work and responsibilities and preferred to be directed". Many people prefer this approach because of the fact that it is only the therefore strategic planning should normally be top-down.
2. Bottom-up approach: in this approach, the various units of an organization prepare their own goals and plans and send them to the top management for approval. The plans and goals when approved become the planning activities of the organization. This approach is based on a theory of human nature which state that "employees like work and responsibilities and are more creative and committed if they participate in the planning and running of an enterprise. In the bottom-up approach, planning start from the lower rank of the organization and planning activities are reviewed and enlarged as they move up in the

organization level until top management is reached

3. Goals down-plans up approach: in this approach, the top management sets corporate goals after looking at the opportunities and requirements. the various units of the organization are responsible for developing plans designed to help the company reach the goals. These plans when approved by the top management become the official plan of the organization

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for the period. This approach is based on participative management principle.

CRITERIA FOR EFFECTIVE PLANNING

1. planning must not be left to chance
2. conscious efforts must be made for planning purposed
3. planning must be organized and controlled
4. planning must have definite and clear focus
5. goals, premises, strategies and policies must be communicated.
6. there must be cognizant support of all concerned about the planning purposes
7. planning must include awareness and acceptance of change
8. there must be adequate resources needed for the planning
9. people involved in planning must be committed to it

STEPS IN PLANNING: to sketch the steps involve in planning and briefly

Environmental scanning

Being aware of
opportunities and threat
in the environment.

Set objectives and Compare planning Implement the plan
goals and objectives

Consider planning

Determine deviation Choose the most
premises

promising alternative

- assumptions

-constraints

- Environmental scanning

Compare alternatives

Identify

- Setalternatives

objectives and goals

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- Consider planning premises

- Identify alternatives
- Compare alternatives
- Choose the most promising alternative
- Implement the plan

For a successful planning, the steps illustrated above should be carried out:

1. Environmental scanning: This gives one the awareness of the opportunities and threat in the environment. when applied to field development planning, environmental scanning is done by SWOT Analysis: strength analysis: sweet oil, high API (i.e low gravity) oil weakness analysis: solution gas, low API (i.e high gravity) oil opportunity analysis: multilateral wells, gas reserves, high oil price threat analysis: vandalization, theft of equipment, community pressure.
2. Set objectives and goals: The purpose of every plan is to facilitate the accomplishment of an organization goals when applied to field development planning: the objectives are economic incentives (maximizing invertment on oil production), domestics needs, natural concerns and global concern. the typical FDP goal is to develop and deplete a hydrocarbon resource with minimum expenditure and environmental impact while approximately recovery, net present volume, and safety. These goals must be achieved within budget at specified time without compromising quality and safety for the FDP to be successful.
3. Consider planning premises: the assumptions and constraints involved in planning must be considered. The constraints would be in form of human resources, fiscal resources and material resources.
4. Identify alternatives: alternative ways/methods of achieving the planned goals must be identified.
5. Compare alternatives: these alternatives must be analyzed and compare with the aim of

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choosing the most promising one.

6. Choose the most promising alternative. The alternative that will result in an optimum accomplishment of the goal should be chosen.
7. Implement the plan
8. Compare planning and objective (and determine the deviation) to monitor plan implementation and develop a feedback mechanism for an improved planning

EXTERNAL FACTORS AFFECTING PROJECT PLANNING

1. Public needs
2. Market need
3. Natural goal
4. Industry stability
5. State of the technology

6. Competition

RESERVES

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DEFINITION OF BASIC CONCEPTS

Oil Field

An oil and gas or petroleum field is an area consisting of a single or multiple hydrocarbon reservoirs grouped or related to the same structural features or stratigraphic conditions

OR An area is defined to be a field by the government or regulating agency.

RESERVOIR

A petroleum reservoir is a porous and permeable formation containing an industrial and separate natural accumulation of producible hydrocarbon or other substances which is confined by and impermeable rock or water barrier and is characterized by a single natural pressure system

RESERVE.

This is the estimated volume of crude oil, condensates, natural gas, liquids and associated marketable substances anticipated to be commercially recoverable and marketable from a given date forward under existing economic conditions by established operating practices and under current government regulations

Recoverable Proved reserves Developed Producing

Hydrocarbon (economical) reserves reserves

resources i.e reserves

Possible reserves

Undeveloped Non-producing

Unrecoverable

Probable reserves reserves reserves

□ Proved reserves: are the estimated quantity of the hydrocarbon which geological and engineering data demonstrate with reasonable certainty to be recoverable in future years from known reservoirs under existing economic, political and operating conditions. Here, the probability factor is around 85% hence economic possibility is supported by either

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actual production or conclusive formation evaluation.

□ Probable reserves: these are less certain than proved reserves and more likely to be recovered than not. They are the estimated quantity of the hydrocarbon which geological and engineering data demonstrate with reasonable probability to be recoverable in future years from known reservoirs under existing economic, political and operating conditions. Here, the probability factor is around 50%.

□ Possible reserves: these are less certain than probable reserves and more likely not to be

recovered. They are the estimated quantity of the likely hydrocarbon which geological and engineering data demonstrate with some possibility to be recoverable in future years from reservoirs under existing economic, political and operating conditions. Here, the probability factor is around 15%.

Proved estimate $\equiv$ conservative proved + probable estimates $\equiv$ realistic

proved + probable + possible estimate $\equiv$ optimistic

□ Developed reserves: these are parts of a field proved reserves that are recovered through existing wells and operating conditions. Here, timing, stimulation and completion are short-time activities whose cost is less than 10 to 20% of the total cost of a new well.

A developed reserves can be classified into producing and non-producing reserves.

Producing reserves: are recovered from existing wells producing from completed intervals under existing operating conditions.

Non-producing reserves: are known to be located behind the casing in the existing well or at mirror depth below the bottom hole of the well and are expected to be produced through these well in the predictable future.

The cost of making such production available is relatively small compared to producing new wells.

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□ Undeveloped reserves: these are parts of a proved reserve that cannot be recovered from existing conditions. Generally, additional wells with new completions will be required in order to recover it.

□ Contingent reserves: these are significant quantities of hydrocarbon resources estimated to be recoverable but are not predictable because of existing economic, political, environmental or technological conditions. Included in this category are resources that are not producible because there are no developmental plan and major capital requirements are required for facilities such as offshore platform, pipeline etc or because of uncertainties related to the terms of lease or production license.

Contingent reserves are categorized as proved, probable or possible based on the same technical guidelines as reserves. If, however, the conditions change to the extent that contingent reserves can be classified as reserves, they should be transferred into their appropriate reserves category.

□ Speculating potential: this represents the quantity of hydrocarbon located in unproven traps, undrilled or deeper reservoirs underlying production fluids where geological conditions are believed to be favourable for hydrocarbon production.

Speculating potential provides estimated of the quantity which if present will eventually be recovered. These potential form the basis for exploration.

METHOD OF RESERVES ESTIMATION

The standard methods for reserves estimation can be employed in either of the two ways namely:

(i) deterministic modelling, and (ii) stochastic/probability modelling. In practice, both deterministic

and stochastic/probability modeling are used; stochastic modeling are incorporated into the deterministic modeling.

Deterministic Modeling

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7758Ah

- Uses exact mathematical equation e.g STOIIP (or N) = Boi
- Advantage: easily understood, easily calculated
- Disadvantages: the modeling most time does not reflect natural situation hence it is not more realistic.

Stochastic/Probability Modeling: this involves probabilities or uncertainties. It uses parameters averaging technique. For example, porosity determination involves breaking the reservoir thickness into depth intervals with different porosities and find the average as given below:

$$\sum h \phi$$

$$\text{Average porosity, } \bar{\phi} = \frac{\sum h \phi}{\sum h}$$

$$\sum h \phi S_w$$

$$\text{Also, average water saturation, } \bar{S}_w = \frac{\sum h \phi S_w}{\sum h \phi}$$

$$\sum h \phi$$

$$\sum h$$

$$\sum h \phi S_w$$

$$\bar{S}_w =$$

$$\frac{\sum h \phi S_w}{\sum h \phi}$$

STANDARD METHODS

- Volumetric method
- Material balance method
- Production decline curve method
- Extrapolating oil cut trends
- Extrapolating observed reservoir trends
- Reservoir simulation trends
- Reservoir simulation technique
- Classical techniques.

The deterministic modeling is fan to undertake and provides “quick look” estimate while the stochastic modeling is believed to be more realistic as it incorporate various uncertainties affecting the estimation of resource volume and reserves.

Standard Methods

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1. Volumetric Methods

a. Area-depth and recovery factor

Reserve = (HCIIP X RF) - Cumulative Production [ultimate recovery = HCIIP X R.F]

Reserve = ultimate Recovery - Cumulative production.

i.e before production, reserve = ultimate Recovery.

b. Net pay volume and recovery factor

c. Volumetric sweep method

$N_r = (N - N_p) \cdot f$ where $N = \text{HCIIP}$

N_r = residual oil volume

f = volumetric sweep efficiency

$7758 \cdot h(1 - \frac{N_p}{N}) \geq 7758 \cdot h \cdot f$

$\frac{N_p}{N} = [1 - f]$

$\frac{N_p}{N} \cdot N = N_p$

For water drive, ΔP is small, $B_{oi} = B_o$

$7758 \cdot h \cdot (\frac{N_p}{N} - \frac{N_{pw}}{N} - \frac{N_{pg}}{N})$

$\frac{N_p}{N} = \frac{N_{pw}}{N} + \frac{N_{pg}}{N}$

$\frac{N_p}{N}$

2. MATERIAL BALANCE METHOD (gives total reserves)

a. calculation method

b. geological techniques (for gas reserves)

3. Production Decline Curve Method (gives developed reserves)

a. exponential decline

b. harmonic decline see example. $Q =$

c. hyperbolic decline here q is reserve (oil)

4. Extrapolating Oil Cut Trends

a. log oil cut vs cumulative oil produced

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b. GOR (or log GOR) vs cumulative oil produced

c. log water cut vs cumulative oil produced

$\frac{Q_w}{Q_o} = \frac{N_{pw}}{N - N_{pw} - N_{pg}}$

note: oil cut, $\frac{Q_w}{Q_o} = \frac{N_{pw}}{N - N_{pw} - N_{pg}}$; $\frac{Q_w}{Q_o} = \frac{N_{pw}}{N - N_{pw} - N_{pg}}$

$\frac{Q_w}{Q_o} = \frac{N_{pw}}{N - N_{pw} - N_{pg}}$

5. Extrapolating Observed Reservoir Trends

a. reservoir pressure against cumulative oil produced

b. movement of oil-water contact or volume of secondary gas cap against cumulative oil production.

Ways of controlling GOR

- produce at low rate (pin down)
- do intermittent production (hydro production)
- gas cycling

6. reservoir simulation techniques

7. classical techniques

Buckle-leverett, wedge method, parton, stiles method.

Note.

- Any method using production data will give developed reserve
- Undeveloped reserve dictates drilling equipment
- Undeveloped reserve = UR - developed reserve.

Extra Note on Production Decline Curve Method: a producing reservoir is a depleting system hence production rate will eventually decline with time. The plot of the decline where there is sufficient data could be used to illustrate past and present data and predict future performance.

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The limitations of this method of analysis is to assume that all factor affecting the performance in the past remain effective throughout the producing life.

The factors that cause changes in the production rate of a well includes reservoir pressure, production, changing production method, workover, well treatment and physical changes around the wellbore, weather and market conditions, efficiency of artificial lift, former reservoir condition such as fw, μ , GOR.

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1

INTEGRATED FIELD

DEVELOPMENT & PLANNING

(PNG 517)

Okologume C. W.

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Course Content

Introduction: [stage-gate process of oil field development, definitions of FDP and the keywords].

□ Stages and sections of FDP; Criteria considered in field development;
SWOT analysis during field development.

□ Conditions for and barriers to field development;
o Methods of improving field development performance.

□Field development cycle.

- Factors affecting options employed in field development.
- Subsurface and surface aspects of FDP.
- Production system options for offshore field development.
- Gas field development.
- Marginal field development.
- Approaches to marginal field development (farm out/farm in arrangement).
- Production system options for marginal field development.
- Management of marginal field.

3

Overall cycle of EP activities

The Upstream industry is characterized by very long cycles which requires long-term vision and decisions.

□ The life cycle of a hydrocarbon field can last for 3 to 5 decades.

□ It is subdivided into several phases separated by a series of GO/no GO steps.

□ These major decision steps are crucial as they are directly related to the following major constraints:

- the reduction of the level of uncertainties related to the uniqueness of each field
- the huge investments tied up in each new phase
- the technological challenges and know-how required to move to the next step.

Ironically, the field is really only fully understood at the end of its life.

4

□ We can subdivide a field's life into three major phases: Exploration, Development, and Production, all of which have specific objectives.

□ Each phase involves numerous competencies that we can classify in several major specific technical "professional disciplines" such as: Geologist, geophysicist, reservoir engineer, driller, petroleum engineer etc....

In addition to these technical domains - like in many other industries - there is also a need for other skills and jobs like economists, lawyers, Human Resources, HSE specialists, etc. ...

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□ The objective of the exploration phase is to

discover hydrocarbons in commercial quantities.

How do we start?

□ With an idea, an initial geological concept generated by geologists and geophysicists - also called “explorationists” - using their experience and studying all available existing data and knowledge.

□ Large regions of the globe need to be evaluated, for many years, trying to reconstruct the geological history of the basins.

Why?

□ To identify potential attractive zones.

□ If successful, this phase results in the application for mining rights for a specific area.

□ This is why we call this part the “Regional evaluation”.

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What comes next?

□ Once the exploration acreage is obtained, detailed studies can be initiated, including the acquisition of new data such as seismic data, outcrop analysis, satellite data, laboratory analysis, etc.

□ This work is done in the part called “exploration”.

The results of all these studies can lead to the definition of a “prospect”, always unique, that will or will not be drilled.

□ Geologists and Geophysicists work together with Drilling Engineers to prepare the drilling campaign.

□ Several exploration wells can be drilled, targeting and testing different objects and geological concepts.

□ This phase, which can last from 2 to 5 years, could lead to a discovery.

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How do we define success?

□ Success in exploration is: finding a significant or economic amount of hydrocarbon.

□ A typical chance of success of an exploration well is around 10 to 20% in under-explored areas.

And just keep in mind the order of magnitude of the cost, of one deep offshore exploration well: around 100 million dollars.

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Cont'd

- Once a discovery is made, another evaluation process starts, dedicated to the appraisal of the size and characteristics of the discovery.
- Several appraisal wells can be drilled.
- A multidisciplinary team composed of geologists, geophysicists, drillers, reservoir engineers will then assess:
 - the extent of the reservoir rock,
 - the volume of recoverable reserves
 - and the production potential.
- This phase can last around 2 to 4 years and ends with the decision to develop the field or not.
- This decision is a very important step as it will be a commitment for huge investments.

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cont'd

- The second major phase after Exploration is “development”.
 - The field development phase is the construction of the field's production facilities, involving all types of professional disciplines but mostly reservoir and petroleum engineers.
 - At the end of this phase, which can last around 3 to 6 years, production can start - this is also called “first oil”.
- Just to give you an order of magnitude, a single offshore Floating Production Surface and Offloading vessel - also called FPSO - could cost around 10 Billion dollars.

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- Finally the third major phase in the life of a field is production.
- A field's typical production life includes:
 - A ramp-up related to the increasing number of production wells being drilled and connected
 - A “plateau” which corresponds to the maximum of the field's production potential
 - And a decline.
- This production phase can last 20 to 30 years but is different for each field.

What happens at the end of the economic production life of the field?

□ The site is restored and/or rehabilitated. This can last 1 to 3 years depending on the size of the installations.

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DEFINITION OF FDP AND THE KEYWORDS

Field Development Planning

□ Field development planning is a multi-disciplinary and integrated approach of performing and analyzing feasibility studies on the exploitation, design basis, engineering, operations and maintenance, and economics aspects of developing an oil field in order to define the optimal development plan and obtain a final investment decision.

□ The multi-disciplinary project team usually needed in field development planning includes geologic and geophysical, engineering, operations and management personnel.

Field development planning explains WHAT will be done whereas the project execution plan explains

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The main steps of an oil or gas field dev. project

A hydrocarbon field has a long life cycle:

□ From the discovery of a petroleum deposit to the first oil/gas, exploration and production activities are spread over several decades.

□ Five main steps can be distinguished in the life of an oil or gas field:

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EXPLORATION AND EVALUATION

1. Field discovery

□ oil and gas are trapped in reservoir rocks buried underground (on shore/off shore),

□ to locate hydrocarbon accumulation, geoscientists analyze images of the earth's

subsurface produced by seismic echography. They build a model of geological layering of the subsurface, and then identify potential reservoirs called “prospects”,

□exploration wells need to be drilled to check if the identified prospect does indeed contain hydrocarbons. These wells are typically several kilometers deep.

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2. Field evaluation

Once discovery has been confirmed, 3D numerical reservoir simulation models are built:

□to estimate the initial volume of oil and gas in the reservoir,

□to simulate the reservoir fluid flow behavior and optimize the field development scenario (number, type and location of wells, level of field production, etc...).

□Appraisal wells are drilled to improve the field description through further data acquisition.

An economic assessment is performed taking into account revenue according to production forecasts and the estimated development costs. If the required economic criteria are met, the field is developed and then produced.

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OIL & GAS PRODUCTION

3. Field development

A field development plan establishes the following:

□the number of wells to be drilled to reach production objectives,

□the recovery techniques to be used to extract the fluids within the reservoir,

- the type and cost of installations, such as platforms, depending on the marine environment (tides, storms, waves, winds, corrosion,
- the separation systems for gas and fluids,
- the treatment systems needed to preserve the environment.

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4. Field production

The time period over which hydrocarbons may be extracted varies between 15 to 30 years and may be extended up to 50 years or more for «giant fields».

□ The lifetime of a reservoir is composed of different successive phases:

- a period of production increase,
- a stabilization phase or «plateau»,
- injection phases (water, gas or chemical products) to «assist» the hydrocarbon recovery and thus maintain a satisfactory volume of produced resources,
- the depletion period when hydrocarbon production declines progressively.

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5. Field abandonment

When the hydrocarbon production rate becomes non economical, the reservoir is abandoned.

□ Before abandoning the field, the oil companies:

- dismantle facilities such as platforms,
- put the well in a safe state,
- preserve the field's residual hydrocarbon reserves of the

field,
☐ clean, depollute and
 rehabilitate the site.

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Marginal Field Development Planning

☐ Marginal field can be defined as discovery (appraisal or not) in the range of 5 to 20 million barrels reserve, that has remained undeveloped for a fairly long period of time (usually 5 to 10 years) and fall on the border line btw. economic and uneconomic reserves hence they are economically sensitive.

A Marginal oil field is:

- ☐ Any oil discovery whose production would, for whatever reasons, fail to match the desired or established rates-of-return of the leaseholder (Egbogah, 2012).
- ☐ Any field that has reserves booked and reported annually to the Department of Petroleum Resources (DPR) and has remained unproduced for a period of over 10 years.

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Cont'd

☐ According to the US Legal.com, marginal field refers to an oil field that may not produce enough net income to make it worth developing at a given time. However, should technical or economic conditions change; such fields may become commercial fields.

☐ Mainly characterized by low reserves and are economically sensitive to develop.

☐ Marginal fields are currently-estimated to contribute about 30% to 40% of global oil produced and about 3 to 4% percent of the total crude production in Nigeria.

☐ Gaining a growing importance due to the natural production decline of large, mature fields.

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Reasons for economic sensitivity of
 marginal field development

- i. Field size to field ranking
- ii. low reserves either as a result of low STOIP or

low recovery factor

iii. unfavourable fiscal regime

iv. marginal economics which could either be as a result of low crude oil price or low gas price or both.

v. low production rate especially in thin reserves

vi. relatively long distance from existing infrastructure

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Reasons cont'd

v. water depth

vi. reservoir rock and fluid characteristics such as low API gravity, high viscosity, low porosity, low permeability k.

vii. high and unfavourable development cost.

(it must be noted that the development of some field that are uneconomical in some parts of the world (such as West Africa) due to high development cost may be profitable in some parts of the world (such as Gulf of Mexico) where they have low development cost.

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Approaches to Marginal Field

Development

□ Several approaches are open to the development of marginal field.

□ the most appropriate development scheme/approach will, however, depend on how innovative the arrangement is and how favourable the prevailing fiscal regime is.

□ The basic approach to the operations of marginal field is the farm out/farm in arrangement

Farm out/farm in arrangement

□ In this type of arrangement, the concessionaire is not interested in drilling at a time but agrees to assign the lease of the field or a portion of it to another operator or farminee who is ready to develop the acreage.

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f/f cont'd

□ He may not retain an interest (royalty) in the

production. This type of arrangement is characterized

by:

i. Interest acquisition: in this case, the farminee carries out an agreed work programme in return for a percentage interest from the field.

□ He will, however, take responsibility for all aspects of development including drilling, operation and finance.

ii. Overriding Royalty: in this case, the farm out party assigns his interest to the farminee who will earn an accruing interest.

□ However, after field development when the farminee must have earned an agreed payment on the investment, the farminee will back in on agreed rate of interest.

iii. Tarriff Payment: in this case, the farminee receives a fixed or adjustable tariff in dollar per barrel. The fee could also come as a percentage of net production.

iv. Others in such as acreage swap, partial relinquishment, transfer of as taken interest in the asset, sublease or

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Production System Options For Marginal Field Development

i. existing production platform

ii. floating production system

iii. mobile offshore production unit (MPU)

iv. subsea completion system

v. small facilities, unmanned structures or light-weight platform

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Management of marginal field

□ The need for higher return on marginal investment forces marginal field operators to look into ways of carrying out petroleum business different from other methods.

□ Some of the new options/alternatives being adopted include:

i. Partnership and allowance concept,

ii. Common sharing of production facilities and

iii. use of total quality management (TQM)

Risk and uncertainties are inherent in petroleum investment.

□ They must be identified and the risk must be quantified.

□ Dealing with risk and uncertainties will affects investment strategy.

- Some of the current fiscal regime that will impart marginal field economy include royalty, signature bonus, production bonus.
 - the property varies from 85% to 75%, investment task credit 10%
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OIL FIELD

- An oil and gas field is an area consisting of a single or multiple hydrocarbon reservoirs grouped or related to the same structural features or stratigraphic condition.
- Or an area defined to be a field by the government or regulating agency.

Reservoir:

- A petroleum reservoir is a porous and permeable formation containing an individual or separate natural accumulation of producible hydrocarbons and other substances which is confined by an impermeable rock or water barrier and is characterized by a single natural pressure system.

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RESERVE

- This is the estimated volume of crude oil, condensates, natural gas, liquids and associated marketable substances anticipated to be commercially recoverable and marketable from a given date forward under existing economic conditions by established operating practices and under current government regulations

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- Here, the probability factor is around 85% hence economic possibility is supported by either actual production or conclusive formation evaluation.

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□ Here, the probability factor is around 50%.

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□ proved + probable estimates $\equiv$ realistic

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because of existing economic, political, environmental or technological conditions.

□ Included in this category are resources that are not producible because

□ there are no developmental plan and major capital requirements are required for facilities such as offshore platform, pipeline etc

□ or because of uncertainties related to the terms of lease or

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Cont'd

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□ this represents the quantity of hydrocarbon located in unproven traps, undrilled or deeper reservoirs underlying production fluids where geological conditions are believed to be favourable for HC production.

□ Speculating potential provides estimation of the quantity which if present will eventually be recovered. These potential

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METHOD OF RESERVES

ESTIMATION

□ The standard methods for reserves estimation can be employed in either of the two ways namely:

- i. deterministic modeling, and
- ii. stochastic/probability modeling.

□ In practice, both deterministic and stochastic/probability modeling are used;

□ stochastic modeling are incorporated into the deterministic modeling.

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Deterministic Modeling

□ Uses exact mathematical equation e.g.

Advantage:

□ easily understood,

□ easily calculated

Disadvantages:

□ the modeling most time does not reflect natural situation hence it is not more realistic.

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Stochastic/Probability Modeling

□ this involves probabilities or uncertainties.

□ It uses parameters averaging technique.

□ For example, porosity determination involves breaking the reservoir thickness into depth intervals with different porosities and find the average as given below:

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STANDARD METHODS

1. Volumetric method
2. Material balance method
3. Production decline curve method
4. Extrapolating oil cut trends
5. Extrapolating observed reservoir trends
6. Reservoir simulation trends
7. Reservoir simulation technique
8. Classical techniques.

□ The deterministic modeling is found to undertake and provides “quick look” estimate while the stochastic modeling is believed to be more realistic as it incorporate various uncertainties affecting the estimation of

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Standard Methods

1. Volumetric Methods
 - a. Area-depth and recovery factor Reserve = (HCIIP X RF) - Cumulative Production [ultimate recovery = HCIIP X R.F]
 - Reserve = ultimate Recovery - Cumulative production.
 - i.e before production, reserve = ultimate Recovery.
 - b. Net pay volume and recovery factor
 - c. Volumetric sweep method

$$Q_{\text{net}} = (Q_{\text{gross}} - Q_{\text{water}}) \times \text{RF}$$

where $N = HCIP$

N_r = residual oil volume

f = volumetric sweep efficiency

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Cont'd

For water drive, ΔP is small, $B_{oi} = B_o$

2. Material Balance Method (gives total reserves)

- a. calculation method
- b. geological techniques (for gas reserves)

40

3. Production Decline Curve Method (gives developed reserves)

- a. exponential decline
- b. harmonic decline see example
- c. hyperbolic decline

4. Extrapolating Oil Cut Trends

- a. log oil cut vs cumulative oil produced
- b. GOR (or log GOR) vs cumulative oil produced
- c. log water cut vs cumulative oil produced

note: oil cut, water cut,

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5. Extrapolating Observed Reservoir Trends

- a. reservoir pressure against cumulative oil produced
- b. movement of oil-water contact or volume of secondary gas cap against cumulative oil production.

Ways of controlling GOR

- i. produce at low rate (pin down)
- ii. do intermittent production (hydro production)
- iii. gas cycling

6. Reservoir simulation techniques

7. Classical techniques

Buckle-leverett, wedge method, parton, stiles method

42

Note

□ Any method using production data will give developed reserve

□ Undeveloped reserve dictates drilling equipment

□ Undeveloped reserve = UR - developed reserve

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Extra Note on Production Decline Curve

Method:

□ A producing reservoir is a depleting system hence production rate will eventually decline with time.

□ The plot of the decline where there's sufficient data could be used to illustrate past and present data and predict future performance.

□ The limitations of this method of analysis is to assume that all factor affecting the performance in the past remain effective throughout the producing life.

□ The factors that cause changes in the production rate of a well includes reservoir pressure, production, changing production method, workover, well treatment and physical changes around the wellbore, weather and market conditions, efficiency of artificial lift, former reservoir condition such as , GOR.

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GOOD ESTIMATES IN A FIELD
DEVELOPMENT PLAN GIVE A
SOLID FOUNDATION FOR DECISION
IN ALL PHASES OF THE
PETROLEUM ACTIVITY

Thank you!